

A Project Report On

mood based music player

Submitted in partial fulfillment of the requirement for the
award of the degree

Bachelor of Computer Application
BCA

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Internal Guide
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Marwadi
University
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Faculty of Computer Applications (FCA)

Certificate

This is to certify that the project work entitled
Mood based music player
submitted in partial fulfillment of the requirement for
the award of the degree of
Bachelor of Computer Application

BCA
of the

Marwadi University

is a result of the bonafide work carried out by
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during the academic year 2025-26

Faculty Guide

HOD

Dean

DECLARATION

I/We hereby declare that this project work entitled **mood based music player** is a record done by me.

I also declare that the matter embodied in this project is genuine work done by me and has not been submitted whether to this University or to any other University / Institute for the fulfillment of the requirement of any course of study.

Place :

Date :

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1.SYNOPSIS

In this project, emotion recognition and music playback are combined. It will continuously use the computer's webcam to scan the user's face. The dominant emotion is identified by processing each frame using the Deep Face library. A matching song is played using Pygame according to the result, which could be joyful, depressed, irate, etc. The goal is to automate music selection in an entertaining and intelligent manner.

2.PREAMBLE

The majority of us listen to music during happy, sad, or even boring moments. What if, without our help, our devices could sense our feelings and play the perfect song? A music player that reads your emotions and selects a song according to your mood is the idea behind this project.

Overview of the Module

There are four modules in the project:

1. Using OpenCV to integrate webcams
2. DeepFace Emotion Recognition
3. Using Pygame to Play Music
4. Keyboard Inputs for User Interaction

3. TECHNICAL OVERVIEW

Hardware specifications:

A computer with a webcam and at least 4 GB of RAM

An Intel i3 or higher processor

Prerequisites for software:

Python 3.8 or higher

Libraries that are required: IDE for DeepFace, Pygame, and

OpenCV: Visual Studio Code or PyCharm

4.DESIGN AND DEVELOPMENT OF THE SYSTEM

A flowchart, use case diagram, and activity diagram are included in the appendix.

GUI design and coding

The user interface is straightforward, utilizing OpenCV to show the camera feed and terminal messages to indicate the detected mood and playback status.

User Webcam



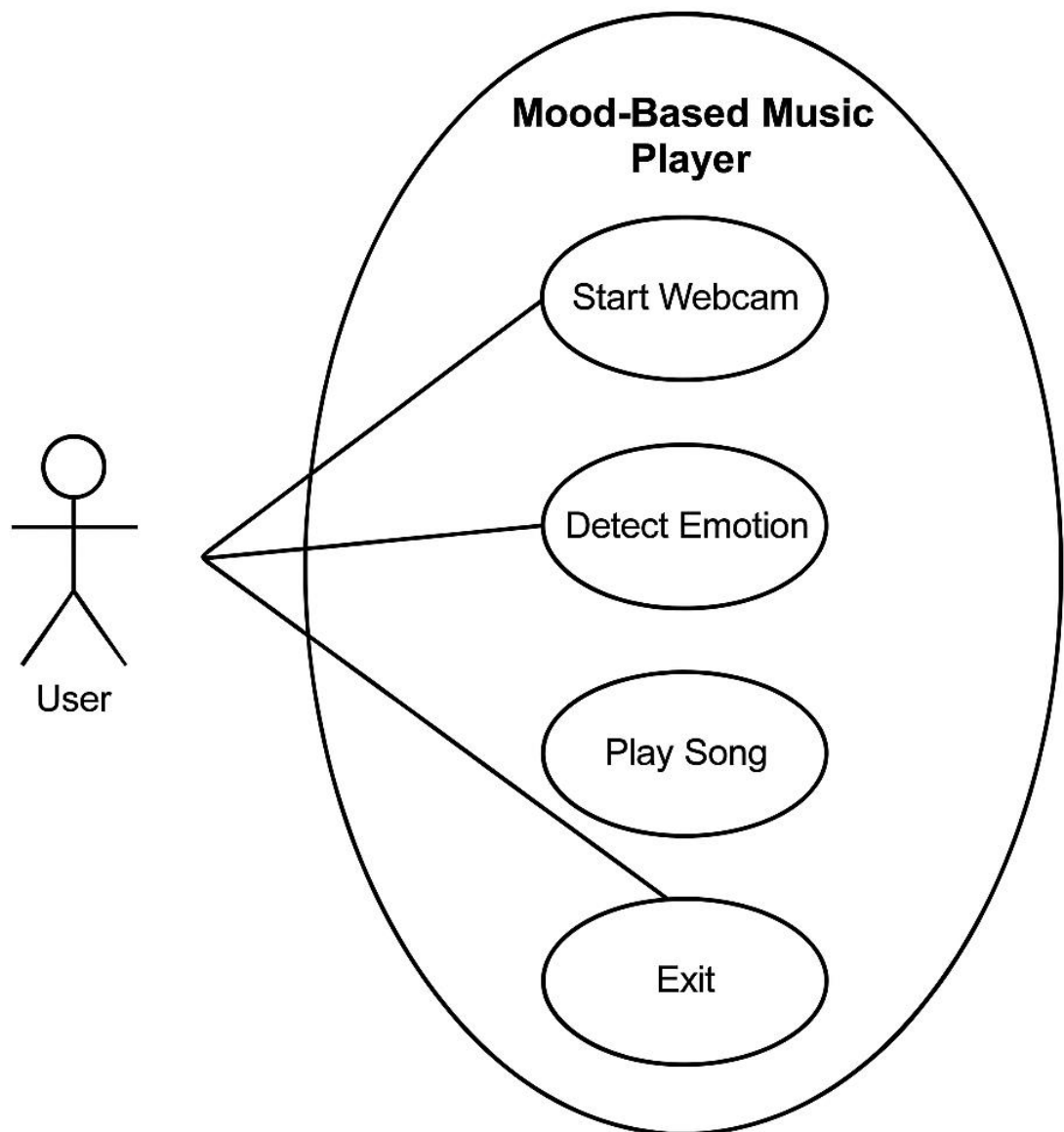
Capture Frame
(cv2)



Analyze Emotion
(DeepFace.analyze)



Play Matching Song
(pygame.mixer)



5.CONCLUSION

The Mood-Based Music Player demonstrates how emotion recognition can improve user experience through music in a way that is highly relatable. Despite having a straightforward interface, it has a strong and engaging feature: it plays music based on your mood.

6.LEARNING DURING SIP

- Hands-on experience using Python libraries such as Deep Face and OpenCV
- Application of deep learning in the real world
Using webcam data in real time
- Debugging and handling errors in real-time systems

7.BIBLIOGRAPHY

- Online References:
 - <https://github.com/serengil/deepface>
 - <https://opencv.org/>
 - <https://www.pygame.org/docs/>
 - <https://docs.python.org>